INCH-POUND

MIL-S-19500/606
10 November 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE ONLY) TRANSISTORS, N-CHANNEL,
SILICON TYPES 2N7291, 2N7293,2N7295, AND 2N7297
JANTXVM, D, R, AND JANSM, D AND R

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an N-Channel, enhancement-mode, MOSFET, radiation hardened (total dose only), power transistor intended for use in high density power switching applications. Two levels of product assurance are provided for each device type specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1, TO-204AA and TO-204AE.

1.3 Maximum ratings. T_A = +25°C unless otherwise specified.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 feet altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7291	150	4	100	100	±20	40.0	25.0	40.0	100	-55	N/A
2N7293			200	200		27.0	17.0	27.0	81	to	N/A
2N7295			250	250		20.0	12.0	20.0	60	+150	N/A
2N7297			500	500		10.0	6.0	10.0	30		500

1/ Derate linearly 1.2 W/°C for T_C > +25°C; $P_T = \dfrac{T_{JM} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\dfrac{T_{JM} - T_C}{\left(R_{\theta JC} \right) \times \left(R_{DS(on)} \text{ at } T_{JM} \right)}}$

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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1.4 Primary electrical characteristics at T_C = 25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc	Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc		$R_{\theta JC}$ max	I_{AS} = I_{DM}	E_{AS} at I_{AS}
				T_J = +25°C at I_{D2}	T_J = +125°C at I_{D2}			
	V dc	V dc Min Max	µA dc	ohm	ohm	°C/W	A(pk)	mJ
2N7291	100	2.0 4.0	25	0.055	0.110	0.83	100	500
2N7293	200			0.10	0.220		81	328
2N7295	250			0.17	0.408		60	180
2N7297	500			0.60	1.56		30	45

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

E_{AS} - Single pulse avalanche energy capability

I_{AS} - Rated avalanche current, non-repetitive

$V_{(ISO)}$ - Source pin to case isolation voltage

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-S-19500 and herein. Where a choice of lead finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.3.2 Internal construction. Multiple chip construction is not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

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4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500, and table V herein.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. (End point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and table IIb herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table VI herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table VI herein.

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with MIL-S-19500 and table IV herein.

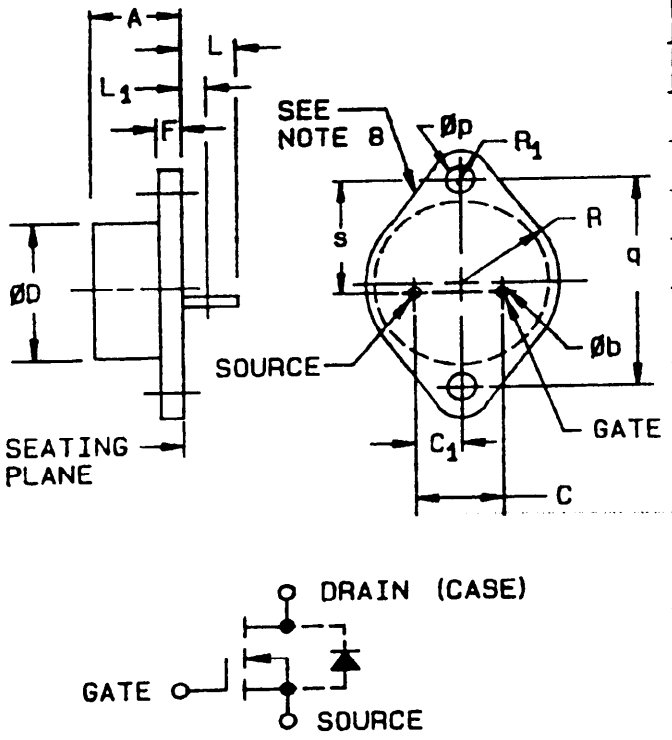
4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ = 0.83°C/W. The following parameter measurements shall apply.

a.	Measuring current (I_M)	10 mA
b.	Drain heating current (I_H)	4 A
c.	Heating time (t_H)	Steady-state (see MIL-STD-750, method 3161 for definition)
d.	Drain-source heating voltage (V_H)	25 V
e.	Measurement time delay (t_{MD})	30 to 60 µs
f.	Sample window time (t_{SW})	10 µs maximum

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.15	
Øb	.057	.063	1.45	1.60	5
Øb	.038	.043	0.97	1.10	6
Øb		.875		22.23	
C	.420	.440	10.67	11.18	1, 7
C_1	.205	.225	5.21	5.72	1, 7
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
Øp	.151	.161	3.84	4.09	
q	1.117	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used, measurements will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. These dimensions pertain to the 2N7291 and 2N7293 types.
6. These dimensions pertain to the 2N7295 and 2N7297 types.
7. Mounting holes shall be deburred on the seating plane side.
8. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions for TO-204.

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Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	MIL-STD-750, method 3161 (see 4.5.3)	MIL-STD-750, method 3161 (see 4.5.3)
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	MIL-STD-750, method 3470, E_{AS} test (see 4.5.4)	MIL-STD-750, method 3470, E_{AS} test (see 4.5.4)
2/	Subgroup 2 of table I herein	Subgroup 2 of table I herein
9	I_{GSS}, I_{DSS1}	Not applicable
10	MIL-STD-750, method 1042 test condition B	MIL-STD-750, method 1042 test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $R_{DS(on)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $R_{DS(on)}$, $V_{GS(th)}$ Subgroup 2 of table I herein
12	MIL-STD-750, method 1042, test condition A	MIL-STD-750, method 1042, condition A or T_A = +175°C and t = 48 hours min 3/
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater. $\Delta R_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ±20nA dc or 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20nA dc or 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10µA dc or 100 percent of initial value, whichever is greater. $\Delta R_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

1/ Shall be performed any time after screen 10.
2/ Shall be performed after E_{AS} test, method 3161, and gate stress test.
3/ Use of this accelerated screening option requires a 1,000 hour life test in accordance with the applicable group E, subgroup 2 life test, and end-points specified herein to be provided to the qualifying activity for review and acceptance.

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4.5.3 Thermal response (ΔV_{SD} measurement). The delta V_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current (I_M) - - - - - - - - - - - - - - - 10 mA

b. Drain heating current (I_H) - - - - - - - - - - - - - 4 A minimum

c. Heating time (t_H) - - - - - - - - - - - - - - - - - - 10 ms

d. Drain-source heating voltage (V_H) - - - - - - - - - 25 V

e. Measurement time delay (t_{MD}) - - - - - - - - - - - 30 to 60 μs

f. Sample window time (t_{SW}) - - - - - - - - - - - - - 10 μs maximum

4.5.4 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$

b. L = .1 mH

c. $E_{AS} = 1/2\ LI_{AS}^2$

d. Initial junction temperature = $+25^{+10}_{-5}$°C

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum

b. t = 250 μs minimum

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc, bias condition C	$V_{(BR)DSS}$			
2N7291				100		V dc
2N7293				200		V dc
2N7295				250		V dc
2N7297				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), Id = Id2	$r_{DS(on)1}$			
2N7291					0.055	Ω
2N7293					0.10	Ω
2N7295					0.17	Ω
2N7297					0.60	Ω
Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), Id = Id1	$r_{DS(on)2}$			
2N7291					0.058	Ω
2N7293					0.105	Ω
2N7295					0.179	Ω
2N7297					0.630	Ω
Forward voltage	4011	Pulsed (see 4.5.1), Id = Id1 V_{GS} = 0 V dc	V_{SD}			
2N7291					1.8	V dc
2N7293					1.8	V dc
2N7295					1.8	V dc
2N7297					1.8	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		±200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1), Id = Id2	$r_{DS(on)3}$			
2N7291					0.110	Ω
2N7293					0.220	Ω
2N7295					0.408	Ω
2N7297					1.56	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)3}$		5.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 10 V dc, R_G = 25Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N7291					170	ns
2N7293					170	ns
2N7295					150	ns
2N7297					160	ns
Rise time			t_r			
2N7291					1120	ns
2N7293					600	ns
2N7295					450	ns
2N7297					260	ns
Turn-off delay time			$t_{d(off)}$			
2N7291					420	ns
2N7293					580	ns
2N7295					650	ns
2N7297					750	ns
Fall time			t_f			
2N7291					380	ns
2N7293					300	ns
2N7295					260	ns
2N7297					180	ns
Subgroup 5						
Safe operating area test	3474	See figure 3 t_p = 10 ms minimum V_{DS} = 80 percent of max rated V_{DS} ($V_{DS} \leq 200$)				
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge		V_{DD} = 0.5 BV_{DSS} $I_D = I_{D1}$ $V_{GS} \leq 20V$ $I_{GS1} = I_{GS2}$	$Q_{g(on)}$			
2N7291					230	nC
2N7293					244	nC
2N7295					252	nC
2N7297					258	nC
Gate to source charge			Q_{gs}			
2N7291					63	nC
2N7293					53	nC
2N7295					48	nC
2N7297					34	nC
Gate to drain charge			Q_{gd}			
2N7291					123	nC
2N7293					125	nC
2N7295					126	nC
2N7297					123	nC
Reverse recovery time	3473	di/dt = 100 A/µs, $V_{DD} \leq 30$ V, Id = Id1				
2N7291			T_{rr}		1400	ns
2N7293					1700	ns
2N7295					2000	ns
2N7297					2300	ns

1/ For sampling plan, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS device.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2 2/		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 3		
Temperature cycling (air to air)	1051	Test condition G, 100 cycles
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7
Decap internal design verification	2075	See 3.3.2
SEM	2077	
Bond strength	2037	Test condition A, all internal wires for each device shall be pulled separately
Die attach integrity	2017	
Subgroup 4		
Intermittent operation life	1042	Condition D, 2,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle t_{on} = 30 seconds minimum
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, 7, and 8

See footnotes at end of table.

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TABLE IIa. Group B inspection for JANS device - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5 2/		
Accelerated steady-state reverse bias	1042	Condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours min. Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA. Read and record I_{DSS} (pre and post) in accordance with table VI.
Electrical measurements		See table VI herein, steps 1, 2, 3, 4, 5, 6, and 7. No more than 15 percent of the sample shall be permitted to have a $\Delta V_{BR(DSS)}$ shift of more than 10 percent and ΔI_{DSS} greater than 50 μA.
Accelerated steady-state gate stress	1042	Condition B, V_{GS} = rated; T_A = +175°C t = 24 hours
Electrical measurements		See table VI herein, steps 1, 2, 3, 4, 5, 6, and 7
Bond strength (Al-Au die interconnects only)	2037	Test condition A, all internal wires for each device shall be pulled separately
Subgroup 6		
Thermal impedance	3161	See 4.5.2

1/ For sampling plan, see MIL-S-19500.
2/ A separate sample may be used for each test.

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TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Temperature cycling (air to air)	1051	Test condition G, 25 cycles
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 3 2/		
Intermittent operation life (LTPD)	1042	Test condition D, 2,000 cycles; a cycle shall be 30 seconds minimum
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, 7, and 8
Bond strength	2037	Test condition A; all internal bond wires for each device shall be pulled separately
Subgroup 4		
Decap internal visual (design verification)	2075	See 3.3.2
Subgroup 5		
Not applicable		
Subgroup 6		
Not applicable		

1/ For sampling plan, see MIL-S-19500.

2/ If group B-3 is to be continued to group C-6, bond strength test may be performed after group C-6.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Thermal shock (glass strain)	1056	Condition B
Terminal strength (tension)	2036	Test condition A, weight = 10 pounds; t = 15 s
Hermetic seal	1071	
Fine leak Gross leak		
Moisture resistance	1021	Omit initial conditioning
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 3		
Shock	2016	
Vibration variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Intermittent operation life	1042	Test condition D, 6,000 cycles; a cycle shall be 30 seconds minimum
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Group D inspection.

Inspection 1/	MIL-STD-750		Symbol	Preirradiation limits		Postirradiation limits		Units
	Method	Conditions		M, D, and R		M, D, and R		
				Min	Max	Min	Max	
Subgroup 1								
N/A								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation	1019	2/ 3/						
End point electricals								
Breakdown voltage, drain to source	3407	V_{GS} = 0; I_D = 1 mA bias cond. C	$V_{BR(DSS)}$					
2N7291				100		100		V dc
2N7293				200		200		V dc
2N7295				250		250		V dc
2N7297				500		500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA	$V_{GS(th)1}$					
2N7291				2.0	4.0	2.0	4.0	V dc
2N7293				2.0	4.0	2.0	4.0	V dc
2N7295				2.0	4.0	2.0	4.0	V dc
2N7297				2.0	4.0	2.0	4.0	V dc
Gate current	3411	V_{GS} = 20 V V_{DS} = 0 bias cond. C	I_{GSSF1}		100		100	nA dc
Gate current	3411	V_{GS} = 20 V V_{DS} = 0 bias cond. C	I_{GSSR1}		-100		-100	nA dc
Drain current	3413	V_{GS} = 0 bias cond. C V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)	I_{DSS1}					
2N7291					25		25	µA dc
2N7293					25		25	µA dc
2N7295					25		25	µA dc
2N7297					25		25	µA dc

See footnotes at end of table.

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TABLE IV. Group D inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Preirradiation limits		Postirradiation limits		Units
	Method	Conditions		M, D, and R		M, D, and R		
				Min	Max	Min	Max	
Subgroup 2 - Continued								
Static drain to source on-state resistance	3421	V_{GS} = 10 V Cond. A pulsed see 4.5.1 $I_D = I_{D2}$	$R_{DS(on)1}$					
2N7291					0.055		0.055	Ω
2N7293					0.100		0.100	Ω
2N7295					0.170		0.170	Ω
2N7297					0.600		0.600	Ω
		V_{GS} = 10 V, cond. A						
2N7291								Ω
2N7293								Ω
2N7295								Ω
2N7297								Ω
Drain source on state voltage	3405	V_{GS} = 10 V $I_D = I_{D1}$ Cond. A pulsed, see 4.5.1 $I_D = I_{D1}$	$V_{DS(on)}$					
2N7291					2.32		2.32	V dc
2N7293					2.70		2.70	V dc
2N7295					3.58		3.58	V dc
2N7297					6.30		6.30	V dc
		V_{GS} = 10 V cond. A						
2N7291								V dc
2N7293								V dc
2N7295								V dc
2N7297								V dc

1/ For sampling plan see MIL-S-19500.

2/ Inspection requires all subgroup 2 (group D) measurements after exposure to both of the following insitu bias conditions:

V_{GS} = 10 V; V_{DS} = 0
V_{GS} = 0 V; V_{DS} = 80 percent of rated V_{DS}

Each bias condition requires a separate total dose sample.

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TABLE V. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	-55°C to +150°C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 2 1/			12 devices c = 0
Steady-state reverse bias	1042	Condition A: 1,000 hours	
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7	
Steady-state gate bias	1042	Condition B: 1,000 hours	
Electrical measurements		See table VI, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
Not applicable			
Subgroup 4			12 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.83C/W maximum. See 4.5.2	

1/ A separate sample for each test shall be pulled.

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TABLE VI. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc; bias condition C	$V_{(BR)DSS}$			
	2N7291				100		V dc
	2N7293				200		V dc
	2N7295				250		V dc
	2N7297				500		V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
3.	Gate current	3411	V_{GS} = +20 and -20 V dc; V_{DS} = 0 V; bias condition C	I_{GGS1}		±100	nA dc
4.	Drain current	3413	V_{GS} = 0 V dc; bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
5.	Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc; condition A pulsed (see 4.5.1) $I_D = I_D2$	$r_{DS(on)1}$			
	2N7291					0.055	Ω
	2N7293					0.100	Ω
	2N7295					0.170	Ω
	2N7297					0.600	Ω
6.	Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc; condition A pulsed (see 4.5.1) $I_D = I_D1$	$r_{DS(on)2}$			
	2N7291					0.058	Ω
	2N7293					0.105	Ω
	2N7295					0.179	Ω
	2N7297					0.630	Ω
7.	Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V dc $I_D = I_D1$	V_{SD}			
	2N7291					1.8	V dc
	2N7293					1.8	V dc
	2N7295					1.8	V dc
	2N7297					1.8	V dc
8.	Thermal response	3161	See 4.5.3	ΔV_{SD}			

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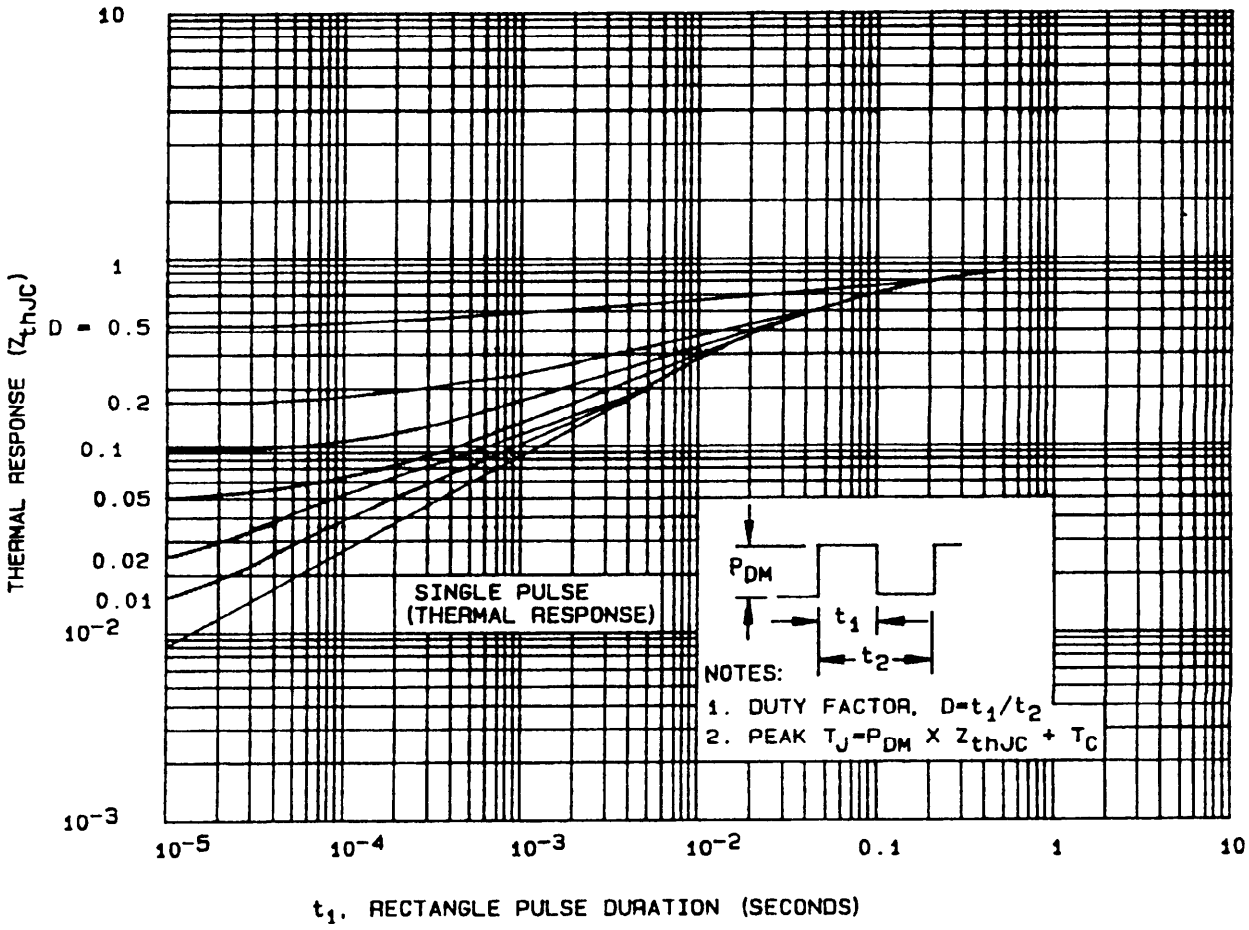


FIGURE 2. Thermal response curves.

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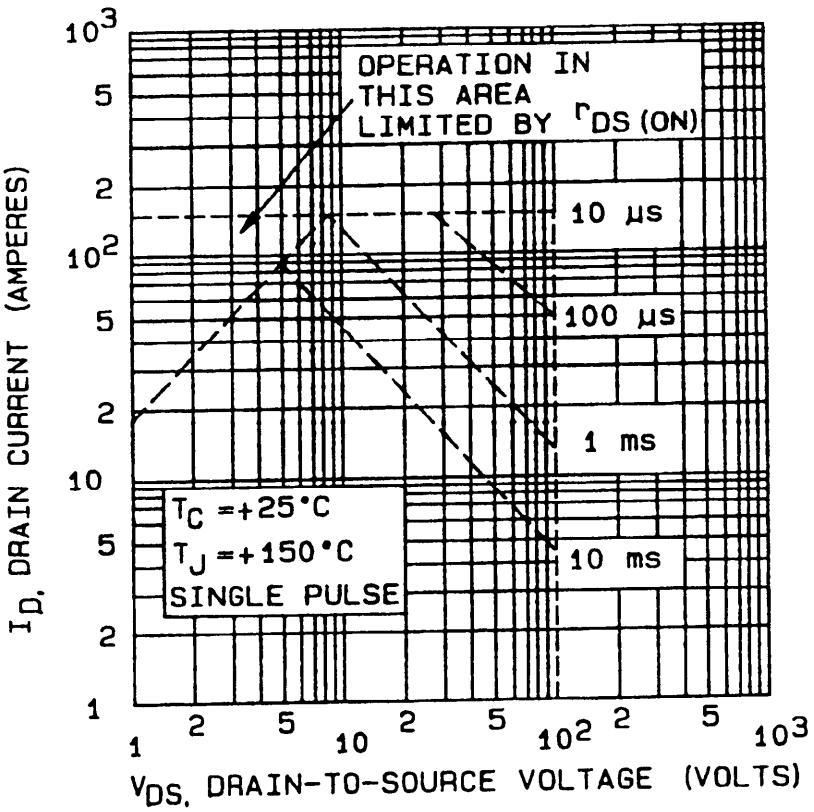


FIGURE 3. Safe operating area graphs.

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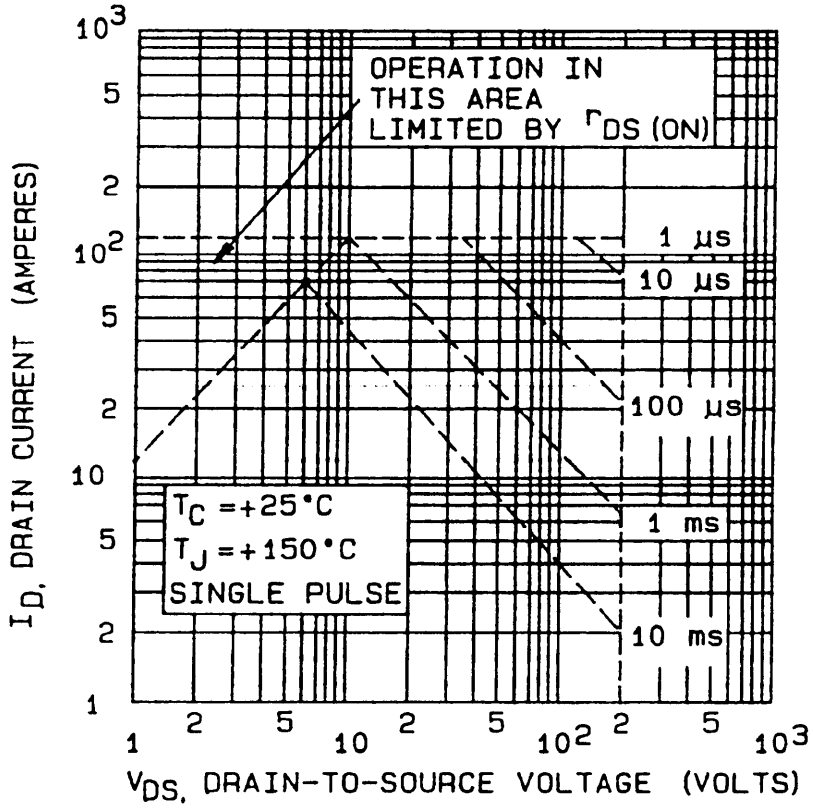


FIGURE 3. Safe operating area graphs - Continued.

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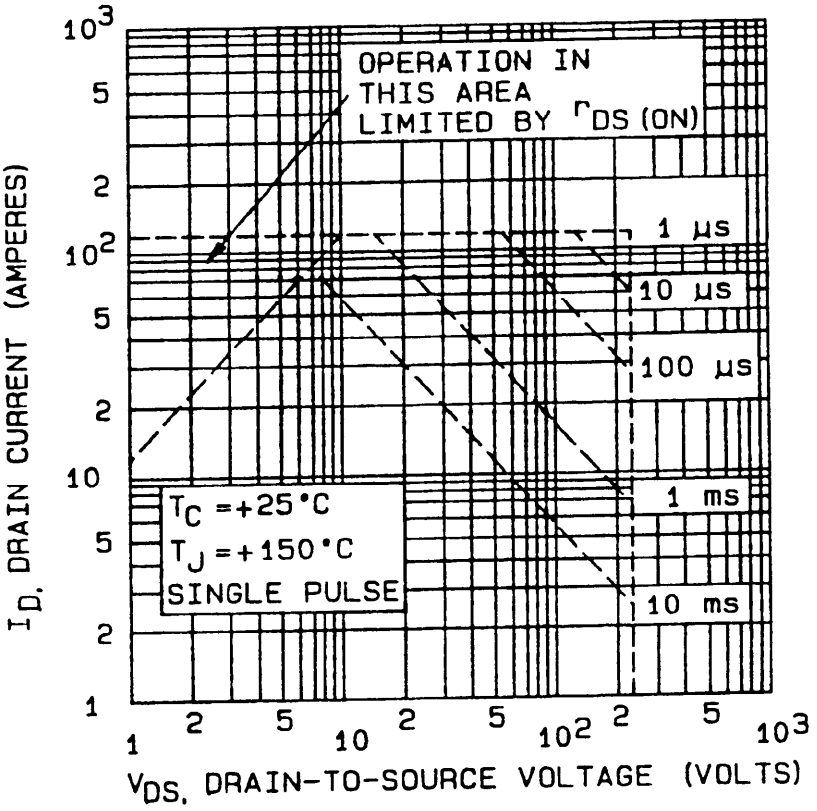


FIGURE 3. Safe operating area graphs - Continued.

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Check the source to verify that this is the current version before use.

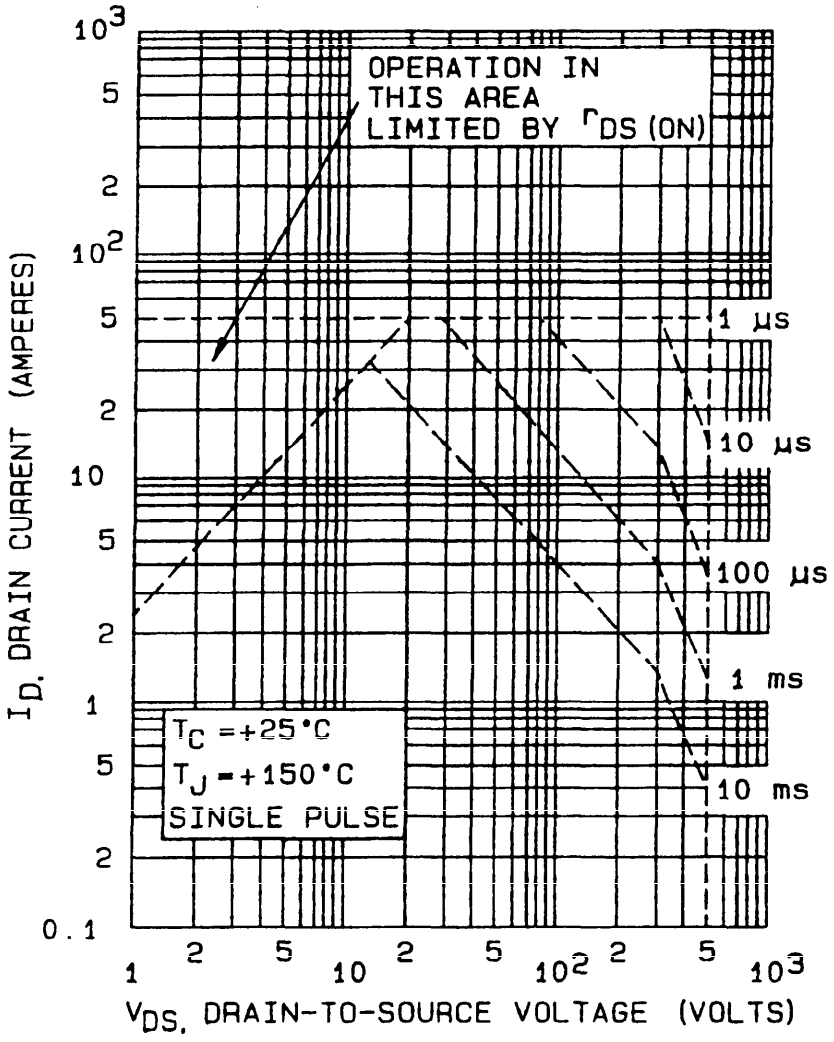


FIGURE 3. Safe operating area graphs - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish may be specified (see 3.3.1).

d. Type designation and product assurance level.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types	Commercial types
2N7291	FRK150 1/
2N7293	FRK250 1/
2N7295	FRK254 1/
2N7297	FRM450 2/

1/ FRKxxxM, FRKxxxD, FRKxxxR, 3 k, 10 k, 100 k RAD(Si)

2/ FRM450M, FRM450D, FRM450R, 3 k, 10 k, 100 k RAD(Si)

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 70, 80
DLA - ES

User activity:
Air Force - 19

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1238)

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